

Punith AJ

Software Engineer | Full Stack | Python | Django | FastAPI | React | AI | Computer Vision

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Professional Summary

Software Engineer with 3.5+ years of experience building enterprise web applications, AI-powered inspection platforms, and LiDAR visualization solutions using React, Django, FastAPI, PostgreSQL, and computer vision technologies.

Professional Experience

Software Engineer | Compviz Technologies Pvt. Ltd. | Dec 2022 – Present

AIMS – Asset Inspection and Monitoring System

- Developed Django backend modules and advanced React components for an AI-powered asset inspection and monitoring platform.
- Built and customized a production LiDAR point-cloud viewer using Potree, integrating backend persistence, engineering measurements, annotations, and visualization tools.
- Developed REST APIs using Django REST Framework with JWT and Session Authentication for project management, annotations, defect management, reporting, and asset inspection.
- Designed PostgreSQL schemas, database models, migrations, and optimized queries for enterprise-scale applications.
- Designed and developed a FastAPI-based AI inference service with persistent model loading, dynamic lifecycle management, and concurrent request handling.
- Designed computer vision training workflows by selecting model architectures and fine-tuning, training, and evaluating YOLO, YOLO-OB, Detectron2, UNet, and SegFormer models using MLflow for experiment tracking and model lifecycle management.
- Integrated raster imagery, contours, and 3D models with point-cloud visualization for engineering analysis.
- Deployed and maintained production systems on Ubuntu using Apache, PostgreSQL, Redis, Celery, and SSL.
- Contributed to an AI-powered asset inspection platform supporting 3,370+ km of transmission lines, 11,000+ towers, 150,000+ tracked defects, and a publicly reported 85% reduction in complaints.

Feeder Analytics Platform

- Integrated Apache Superset with a Django enterprise application.
- Developed backend APIs, administration modules, custom reporting, datasets, and analytics workflows.
- Implemented row-level security, role-based permissions, and secure business intelligence reporting.

Education

M.Tech in Digital Communication – RV College of Engineering, Bengaluru (CGPA: 8.05)

Technical Skills

- **Languages:** Python, JavaScript, SQL
- **Frontend:** React.js, React Native, HTML, CSS
- **Backend:** Django, DRF, FastAPI, REST APIs, JWT
- **Database:** PostgreSQL
- **AI/CV:** Machine Learning, Deep Learning, Computer Vision, NLP, GANs, Pytorch, TensorFlow, NumPy, Pandas, Scikit-Learn, OpenCV, YOLO, YOLO-OB, Detectron2, UNet, SegFormer, MLflow, LangChain
- **Visualization:** Potree, Apache Superset
- **Infrastructure:** Redis, Celery, Ubuntu, Apache, Git